

AI-Generated Text vs Human-Written Text: Can Machines Tell the Difference?

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Abstract

AI writing tools like ChatGPT are everywhere now, and it is getting harder to tell whether something was written by a real person or generated by a machine. In this project, we trained five different models to detect AI-generated text across three datasets: HC3 (medical, legal, and finance answers), NYT Articles (news), and RAID (a hard multi-domain adversarial benchmark from ACL 2024). Our best model, fine-tuned DistilBERT, reached 98.2% on HC3 and 83.0% on RAID. We also found that GPT-3.5 without any training scored just 33.5% on HC3, which is worse than random guessing. One of the most important parts of our project was discovering data leakage in all three datasets, which helped us understand why some accuracy numbers were inflated and what honest performance actually looks like.

1 Introduction

AI writing is everywhere now. Tools like ChatGPT can write a medical explanation, a legal document, or a full news article in seconds, and the result often sounds just as convincing as something a real person wrote. This is starting to cause serious problems. Students are submitting AI-written essays as their own work, fake medical advice is spreading online, and it is getting harder to trust the authenticity of content we read every day.

Being able to automatically detect AI-generated text would help in many ways. Schools could use it to check for academic dishonesty. News platforms could use it to flag AI-generated misinformation. Medical and legal websites could use it to verify that advice is coming from real experts. The need for reliable detection is growing as AI writing tools become more capable and more widely used.

The problem is harder than it might seem at first. Modern LLMs like ChatGPT produce text that is fluent, well-structured, and topically accurate. In many cases even human readers cannot reliably

tell the difference. Previous work has shown that machine learning models can achieve high accuracy on simple in-domain benchmarks (Ippolito et al., 2020), but that performance drops significantly when the domain changes or when adversarial attacks are applied (Dugan et al., 2024). A key concern in this area is whether high accuracy reflects genuine detection ability or simply exploiting artifacts in the dataset, such as formatting differences or scraped metadata.

In this project we wanted to address two questions. First, can machine learning models reliably detect AI-generated text across multiple domains? Second, when a model gets high accuracy, is it learning real linguistic differences or just finding shortcuts in the data? We trained five models ranging from Naive Bayes to fine-tuned DistilBERT and GPT-3.5 zero-shot prompting, and we evaluated them across three datasets of increasing difficulty. We also ran a careful leakage analysis on each dataset to understand what the models were actually learning.

2 Methodology

We treated this as a binary classification problem. The model reads a text sample and predicts whether it was written by a human (label 0) or generated by an AI model (label 1). We trained and evaluated five models.

Naive Bayes. Our simplest baseline uses a bag-of-words representation built with CountVectorizer (5,000 features, English stopwords removed) and a Multinomial Naive Bayes classifier. This model counts how often each word appears and uses those counts to make predictions. It does not understand word order or meaning at all, but it is fast and easy to interpret. We used this as our starting point to see what even a very basic model could do.

Logistic Regression. Our main baseline uses TF-IDF features, which measure how important a word is in a document compared to the rest of the dataset. We used 5,000 features with English stopwords removed and trained with L2 regularization (`max_iter=1000`). One advantage of this model is that the learned weights tell us which words most strongly predict each class, which was useful for our leakage investigation.

Linear SVM. This model also uses TF-IDF features but finds the maximum margin boundary between the two classes. We used `LinearSVC` with `max_iter=2000`. Linear support vector machines work well when there are many features relative to the number of samples, which is typical in text classification. This was our strongest classical model overall.

DistilBERT (Fine-tuned). DistilBERT (Sanh et al., 2019) is a smaller and faster version of BERT that keeps 97% of its performance. Unlike the other models that just count word frequencies, DistilBERT actually understands the meaning and context of words. For example, it knows that the word “run” means something different in a jogging article versus a coding tutorial. We fine-tuned the `distilbert-base-uncased` checkpoint for 3 epochs using AdamW with a learning rate of $2e-5$, batch size 32, and maximum sequence length 256 tokens. We ran this on a Google Colab A100 GPU, which took about 10 to 15 minutes per dataset.

GPT-3.5 Zero-Shot Prompting. Instead of training a model, we used the OpenAI API to send each text sample to GPT-3.5 with a prompt asking it to classify the text as HUMAN or AI. We used `temperature=0` for consistent outputs and `max_tokens=5` since we only needed one word back. No labeled examples were given in the prompt, which is what makes this zero-shot. This approach tests a fundamentally different question: can a large language model detect AI text just by using what it already knows, without any task-specific training?

3 Data

We used three datasets from very different domains. Each was balanced to 5,000 human and 5,000 AI samples for a total of 10,000 per dataset. We used an 80/20 stratified train/test split with `random_state=42` to ensure reproducibility and class balance.

HC3 (Medical / Legal / Finance). The Human ChatGPT Comparison Corpus (Guo et al., 2023) was specifically designed to compare human expert answers with ChatGPT answers on the same questions. Human answers come from real doctors, lawyers, and financial analysts. ChatGPT answers respond to the exact same questions. We loaded the medicine, finance, and `open_qa` subsets directly from HuggingFace using the identifier `Hello-SimpleAI/HC3`. Before balancing, the full loaded dataset contained 84,392 samples across all subsets. After balancing we used 10,000 samples (5,000 per class). This dataset has a complexity rating of 6/10 because expert human writing and ChatGPT answers are both reasonably formal, making the gap smaller than in essay-based datasets.

NYT Articles (News). This dataset (?) pairs real New York Times articles with AI-generated rewrites produced by six different models: GPT-4o, LLaMA-8B, Mistral-7B, Gemma-2-9B, Qwen-2-72B, and Yi-Large. The original data is stored in wide format with one row per article and each AI version as a separate column. We reshaped it into long format with one text per row, yielding 51,168 total samples. After balancing we used 10,000 samples. We gave this a complexity rating of 8/10 because professional journalism and AI rewrites of the same article are stylistically very similar.

RAID (Multi-domain Adversarial). RAID (Dugan et al., 2024) is the most challenging dataset we used. It was published at ACL 2024 specifically as a benchmark to test the limits of AI text detectors. The full dataset contains over 5.6 million samples generated by 11 different AI models including ChatGPT, GPT-4, GPT-2, Mistral, LLaMA, Cohere, and MPT, across 8 completely different domains: news, books, Reddit posts, poetry, recipes, Wikipedia articles, ArXiv academic abstracts, and movie reviews. It also includes adversarial attacks such as paraphrasing and synonym swapping designed to make AI text look more human. We loaded the full training split from HuggingFace and sampled 625 human and 625 AI examples from each of the 8 domains to get a balanced and diverse working set of 10,000 samples. We rated this 10/10 for complexity.

Preprocessing. For all three datasets we dropped null and duplicate rows and kept only the text and label columns. For TF-IDF and `CountVectorizer` models, we always fit the vectorizer on the train-

ing set only and then used it to transform the test set. This is important because fitting on the full dataset would expose the model to test set vocabulary statistics, which is a form of leakage. For DistilBERT we used WordPiece tokenization with padding and truncation at 256 tokens.

Leakage Analysis. One of the most important things we did was check what our models were actually learning by inspecting the top TF-IDF coefficient words from logistic regression. We found issues in all three datasets.

In HC3, human answers mostly came from the Reddit ELI5 subreddit, where people write casually and informally using words like *just*, *really*, *basically*, and *probably*. ChatGPT answers on the other hand are always formal and structured, using words like *important*, *including*, *financial*, and *helps*. The model learned to detect Reddit-style casual writing rather than true authorship differences. This is a domain mismatch problem, not real detection.

In NYT, the top words predicting human included *slideshow*, *controls*, *navigation*, and *mr*. These are HTML artifacts that came from scraping the New York Times website. The model learned that website navigation elements equal human text. This is why NYT accuracy reached 99% and the number is not meaningful.

In RAID, the top AI-predicting words included *ingredients*, *recipe*, *instructions*, and *enjoy*, while top human words included *tablespoons*, *add*, and *pan*. The model learned recipe-domain vocabulary rather than AI writing style. RAID gave the most honest results of the three datasets because the adversarial attacks and domain diversity made it much harder to exploit these kinds of shortcuts.

4 Evaluation

We evaluated all models on a held-out test set of 2,000 samples per dataset (1,000 human and 1,000 AI). We report accuracy, precision, recall, and macro F1-score. We use macro F1 as our primary metric because it treats both classes equally and is more informative than accuracy alone for balanced datasets. All evaluation was fully automated and required no human annotation.

For GPT-3.5, we only ran evaluation on 200 samples per dataset (600 total across all three datasets) because of API cost constraints. The OpenAI API charges per token used, and our 600 calls already cost around \$10. Running on the full 2,000 samples per dataset would have cost approximately \$50,

which was not feasible for us as students. Given that GPT-3.5 scored as low as 33.5% on HC3, well below random chance of 50%, we are confident the conclusion holds regardless of the smaller sample size.

We used Naive Bayes as our simplest baseline, Logistic Regression as our main baseline, Linear SVM as our best classical model, DistilBERT as our best overall model, and GPT-3.5 zero-shot as our LLM prompting approach. The same test set was used for all five models on each dataset so that all comparisons are fair.

5 Results

Table 1 shows accuracy for all five models across all three datasets. Table 2 shows macro F1 scores.

Model	HC3	NYT	RAID
Naive Bayes	84.1	96.5	68.5
Logistic Reg.	90.0	98.7	75.6
Linear SVM	90.0	99.2	74.8
DistilBERT	98.2	99.7	83.0
GPT-3.5 0-shot	33.5	45.5	50.3

Table 1: Accuracy (%) on the test set (2,000 samples per dataset, 1,000 per class). NYT numbers are inflated by HTML leakage. RAID gives the most honest results. GPT-3.5 was tested on 200 samples per dataset due to cost.

Model	HC3	NYT	RAID
Naive Bayes	0.84	0.97	0.68
Logistic Reg.	0.90	0.99	0.76
Linear SVM	0.90	0.99	0.75
DistilBERT	0.98	1.00	0.83
GPT-3.5 0-shot	0.33	0.40	0.50

Table 2: Macro F1 scores across all models and datasets.

DistilBERT was the best model on every dataset. The gap between DistilBERT and GPT-3.5 zero-shot is large: 64.7 percentage points on HC3, 54.2 on NYT, and 32.7 on RAID. This shows clearly that fine-tuning on labeled examples is much more effective than zero-shot prompting for this task, regardless of how powerful the LLM is.

The most surprising result was GPT-3.5 zero-shot scoring 33.5% on HC3. This is significantly below random chance (50%), meaning the model was actively predicting the wrong class more often than not. On RAID it scored 50.3%, essentially the same as flipping a coin. This finding suggests that LLMs are not aware of their own writing style in a way that would let them detect it without training.

RAID was the hardest dataset by a clear margin. Every model dropped substantially compared to HC3 and NYT. Naive Bayes went from 84.1% on HC3 down to 68.5% on RAID. DistilBERT dropped from 98.2% to 83.0%. This consistent pattern across all five models confirms that RAID is a genuinely challenging benchmark that prevents easy exploitation of domain shortcuts.

The confusion matrix for Linear SVM on RAID showed 504 errors out of 2,000 predictions, with 248 human texts classified as AI and 256 AI texts classified as human. This near-equal error distribution suggests the model did not have a strong bias toward either class on this dataset, which is a sign of a harder and more honest evaluation.

6 Discussion

This project taught us a lot, and not just about AI text detection. The most important lesson was that high accuracy does not always mean the model learned the right thing.

Things that went well include building a complete end-to-end pipeline with five models and three datasets, identifying leakage in all three datasets through feature inspection, and finding that GPT-3.5 cannot self-detect without training, which is itself a meaningful research finding. DistilBERT also showed clear and consistent improvement over all classical baselines, especially on the harder RAID benchmark where contextual understanding matters more.

Things that did not go well include the NYT HTML leakage making that dataset unreliable, the domain mismatch in HC3 between Reddit posts and ChatGPT answers, and recipe vocabulary dominating RAID predictions. The GPT-3.5 API also cost much more than we expected for a relatively small number of samples, which limited how much we could test.

If we did this project again, we would spend more time cleaning each dataset before training. For NYT that means stripping HTML artifacts. For HC3 that means using only professional domain answers rather than Reddit posts. We would also run cross-domain experiments, training on one dataset and testing on another, to really measure whether the models generalize or just memorize domain-specific shortcuts.

Future work. With more time, we would focus on six things. First, fixing the NYT leakage by removing HTML text before training to see what the

real accuracy is. Second, cross-domain evaluation by training on HC3 and testing on RAID. Third, upgrading from DistilBERT to RoBERTa-base, which may perform better on RAID. Fourth, testing few-shot GPT prompting by providing labeled examples in the prompt to see if that fixes the zero-shot failure. Fifth, adversarial testing by asking ChatGPT to write text specifically designed to sound human, and checking whether our classifier can still catch it. Sixth, using language model perplexity as an additional feature, since AI-generated text tends to have lower perplexity than human-written text.

Limitations

Our main limitations are dataset leakage in NYT and HC3, the small sample size for GPT-3.5 evaluation due to API cost constraints, and not being able to run cross-domain generalization tests within our project timeline. The NYT accuracy of 99% should not be taken as a real result. We consider RAID to be our most honest and reliable benchmark.

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